

9. The _____ search algorithm repeatedly divides the portion of an array being searched in half.
10. The _____ search algorithm is adequate for small arrays but not large arrays.
11. The _____ search algorithm requires that the array's contents be sorted.
12. If an array is sorted in _____ order, the values are stored from lowest to highest.
13. If an array is sorted in _____ order, the values are stored from highest to lowest.

True or False

14. T F If data are sorted in ascending order, it means they are ordered from lowest value to highest value.
15. T F If data are sorted in descending order, it means they are ordered from lowest value to highest value.
16. T F The *average* number of comparisons performed by the linear search on an array of N elements is N/2 (assuming the search values are consistently found).
17. T F The *maximum* number of comparisons performed by the linear search on an array of N elements is N/2 (assuming the search values are consistently found).
18. Complete the following table calculating the average and maximum number of comparisons the linear search will perform, and the maximum number of comparisons the binary search will perform.

Array Size →	50 Elements	500 Elements	10,000 Elements	100,000 Elements	10,000,000 Elements
Linear Search (Average Comparisons)					
Linear Search (Maximum Comparisons)					
Binary Search (Maximum Comparisons)					

Programming Challenges

1. Charge Account Validation

Write a program that lets the user enter a charge account number. The program should determine if the number is valid by checking for it in the following list:

```
5658845  4520125  7895122  8777541  8451277  1302850
8080152  4562555  5552012  5050552  7825877  1250255
1005231  6545231  3852085  7576651  7881200  4581002
```

The list of numbers above should be initialized in a single-dimensional array. A simple linear search should be used to locate the number entered by the user. If the user enters a number that is in the array, the program should display a message saying that the number is valid. If the user enters a number that is not in the array, the program should display a message indicating that the number is invalid.

2. Lottery Winners

A lottery ticket buyer purchases 10 tickets a week, always playing the same 10 5-digit “lucky” combinations. Write a program that initializes an array or a vector with these numbers and then lets the player enter this week’s winning 5-digit number. The program should perform a linear search through the list of the player’s numbers and report whether or not one of the tickets is a winner this week. Here are the numbers:

```
13579  26791  26792  33445  55555
62483  77777  79422  85647  93121
```

3. Lottery Winners Modification

Modify the program you wrote for Programming Challenge 2 (Lottery Winners) so it performs a binary search instead of a linear search.

4. Charge Account Validation Modification

Modify the program you wrote for Problem 1 (Charge Account Validation) so it performs a binary search to locate valid account numbers. Use the selection sort algorithm to sort the array before the binary search is performed.

5. Rainfall Statistics Modification

Modify the Rainfall Statistics program you wrote for Programming Challenge 2 of Chapter 7. The program should display a list of months, sorted in order of rainfall, from highest to lowest.

6. String Selection Sort

Modify the `selectionSort` function presented in this chapter so it sorts an array of strings instead of an array of ints. Test the function with a driver program. Use Program 8-8 as a skeleton to complete.



VideoNote
Solving the
Charge
Account
Validation
Modification
Problem

Program 8-8

```
#include <iostream>
#include <string>
using namespace std;

int main()
{
    const int NUM_NAMES = 20;
    string names[NUM_NAMES] = {"Collins, Bill", "Smith, Bart", "Allen, Jim",
                                "Griffin, Jim", "Stamey, Marty", "Rose, Geri",
                                "Taylor, Terri", "Johnson, Jill",
                                "Allison, Jeff", "Looney, Joe", "Wolfe, Bill",
                                "James, Jean", "Weaver, Jim", "Pore, Bob",
                                "Rutherford, Greg", "Javens, Renee",
                                "Harrison, Rose", "Setzer, Cathy",
                                "Pike, Gordon", "Holland, Beth" };

    // Insert your code to complete this program

    return 0;
}
```

7. Binary String Search

Modify the `binarySearch` function presented in this chapter so it searches an array of strings instead of an array of `ints`. Test the function with a driver program. Use Program 8-8 as a skeleton to complete. (The array must be sorted before the binary search will work.)

8. Search Benchmarks

Write a program that has an array of at least 20 integers. It should call a function that uses the linear search algorithm to locate one of the values. The function should keep a count of the number of comparisons it makes until it finds the value. The program then should call a function that uses the binary search algorithm to locate the same value. It should also keep count of the number of comparisons it makes. Display these values on the screen.

9. Sorting Benchmarks

Write a program that uses two identical arrays of at least 20 integers. It should call a function that uses the bubble sort algorithm to sort one of the arrays in ascending order. The function should keep a count of the number of exchanges it makes. The program then should call a function that uses the selection sort algorithm to sort the other array. It should also keep count of the number of exchanges it makes. Display these values on the screen.

10. Sorting Orders

Write a program that uses two identical arrays of just eight integers. It should display the contents of the first array, then call a function to sort the array using an ascending order bubble sort modified to print out the array contents after each pass of the sort. Next, the program should display the contents of the second array, then call a function to sort the array using an ascending order selection sort modified to print out the array contents after each pass of the sort.

11. Using Files—String Selection Sort Modification

Modify the program you wrote for Programming Challenge 6 so it reads in 20 strings from a file. The data can be found in the `names.txt` file.